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Platonin mitigates vascular hyporeactivity of thoracic aorta in septic rats

Tsui-Chin Peng, MD,^{a,b,1} Chao-Yuan Chang, PhD,^{c,1} I-Tao Huang, MBBS,^d
Ming-Chang Kao, MD, PhD,^{a,b} Ya-Ying Chang, MD,^a
and Chun-Jen Huang, MD, PhD^{c,e,*}

^aDepartment of Anesthesiology, Taipei Tzu Chi Hospital, Taipei, Taiwan

^bSchool of Medicine, Tzu Chi University, Hualien, Taiwan

^cDepartment of Anesthesiology, Wang Fang Hospital, Taipei Medical University, Taipei, Taiwan

^dSchool of Medicine, The University of Queensland, Brisbane, St. Lucia, Queensland, Australia

^eGraduate Institute of Clinical Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

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ABSTRACT

Background: Vascular hyporeactivity contributes to hemodynamic alterations and circulatory failure in severe sepsis. Among the identified mechanisms, inflammation and oxidative stress are the most crucial ones in mediating the development of vascular hyporeactivity induced by sepsis. Platonin, a photosensitive dye and an antioxidant, possesses potent antiinflammation effects. We elucidated whether platonin could mitigate vascular hyporeactivity of thoracic aorta in septic rats.

Material and methods: Adult male Sprague–Dawley rats were randomized to receive sham operation (Sham), Sham plus platonin (100 µg/kg), cecal ligation and puncture (CLP), or CLP plus platonin (10, 50, or 100 µg/kg) and designated as the Sham, P, CLP, CLP + P(10), CLP + P(50), and CLP + P(100) group, respectively ($n = 6$ in each group). After maintaining for 12 hours, surviving rats were euthanized and thoracic aorta was isolated and vascular reactivity of aortic rings was determined.

Results: Vascular reactivity induced by vasoconstrictors phenylephrine and angiotensin II of the Sham and the P groups ($n = 6$ in both groups) were similar, whereas vascular reactivity of the CLP group ($n = 5$) were significantly lower than those of the Sham group (both $P < 0.001$). Of note, vascular reactivity induced by phenylephrine and angiotensin II of the CLP + P(10) group ($n = 5$) and the CLP group were not significantly different. In contrast, vascular reactivity induced by phenylephrine and angiotensin II of the CLP + P(50) and the CLP + P(100) groups ($n = 6$ in both groups) were significantly higher than those of the CLP group (phenylephrine: $P = 0.024$ and 0.017 ; angiotensin II: $P = 0.031$ and 0.036).

Conclusion: Platonin could mitigate vascular hyporeactivity of thoracic aorta in septic rats.

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* Corresponding author. Department of Anesthesiology, Wang Fang Hospital, 111, Sec. 3, Xinglong Rd., Wenshan Dist., Taipei 116, Taiwan. Tel.: +886 2 29307930x2160; fax: +886 2 29335221.

E-mail address: huangcj1112@gmail.com (C.-J. Huang).

¹ T.-C.P. and C.-Y.C. contributed equally to this study.

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Introduction

Platonic, a photosensitive dye, possesses potent anti-oxidation and antiinflammation effects.¹ Clinical and experimental data have confirmed the beneficial effects of platonic against conditions associated with oxidative stress and/or inflammation.²⁻⁶ For instance, platonic can improve clinical symptoms, mitigate inflammation severity, and maintain the remission state in cases with juvenile rheumatoid arthritis.² Experimental data also confirmed that platonic can improve circulatory failure, attenuate organ injuries, and decrease mortality in septic rats.^{3,4} In addition, the injuries imposed by hemorrhage or ischemia-reperfusion can also be attenuated by platonic.^{5,6}

As mentioned previously, platonic improves circulatory failure in sepsis.³ Vascular hyporeactivity contributes to hemodynamic alterations and circulatory failure in sepsis.⁷ Inflammation and oxidative stress are two crucial mechanisms in mediating the development of vascular hyporeactivity induced by sepsis.⁸ In line with this notion, we thus speculated that platonic may very likely mitigate vascular hyporeactivity induced by sepsis.

To elucidate further, we thus conducted this rodent study with the hypothesis that platonic could mitigate thoracic aorta hyporeactivity in septic rats. This study employed a widely used cecal ligation and puncture (CLP) polymicrobial sepsis model⁹ to facilitate investigation. Vascular reactivity was measured using an isolated thoracic aortic ring platform, as previously reported.¹⁰ In addition, the levels of inflammation and oxidation of the isolated thoracic aorta were also measured.

Methods

The animal study protocol was approved by the Animal Use and Care Committee of Taipei Tzu Chi Hospital, Taipei, Taiwan (105-IACUC-013). Care and handling of the animals were in accordance with the National Institutes of Health guidelines. A total of 36 adult male Sprague–Dawley rats (BioLASCO, Taipei, Taiwan; 200–250 gm) were used in this study.

Sepsis model

As aforementioned, this study employed the CLP polymicrobial sepsis model to facilitate investigation. CLP was performed according to previous protocols.⁹ In brief, all rats were anesthetized with a ketamine/xylazine mixture (110/10 mg/kg body weight, ipf) and one 1-cm transverse laparotomy was performed at the right lower quarter of the abdominal wall sterilely. Then, the cecum was ligated, and two 0.5-cm blade incisions were made followed by abdominal wall closure. To control the effects of operational procedures, another set of rats received laparotomy, cecum identification, and wound closure, but not CLP (i.e. sham operation). To alleviate wound pain, subcutaneous injection of 0.25% bupivacaine (AstraZeneca, Taipei, Taiwan) was performed before wound closure. After recovery, all rats were closely monitored for 12 hours without restraint.

Experimental protocol

Rats were randomized to receive sham operation (Sham), Sham plus platonic (100 µg/kg, Kankohsha Co, Osaka, Japan; dissolved in a mixture of 0.5 mL normal saline), CLP, or CLP plus platonic (10, 50, or 100 µg/kg) and designated as the Sham, P, CLP, CLP + P(10), CLP + P(50), and CLP + P(100) group, respectively ($n = 6$ in each group). Platonic was injected via tail vein immediately after CLP or at the comparable time point in the P group. The dosages of platonic were determined according to a previous report.³ To control for the effects of vehicle, rats of the Sham and the CLP groups also received injection of 0.5-mL normal saline via tail vein. At 12 hours after intervention, all surviving rats were euthanized with decapitation.

In vitro assessment of vascular reactivity

After euthanization, thoracotomy was performed and the descending thoracic aorta was isolated and the connective tissues were cleaned. The distal portion of the descending aorta was snap frozen in liquid nitrogen and stored at -80°C for subsequent analysis. The proximal portion of the descending aorta was cut into ring-like segments (3–4 mm) and then mounted in tissue bath chambers (Radnoti Four Unit Tissue Organ Bath System, ADInstruments Inc. Colorado Springs, CO) containing Krebs solution (NaCl 115.3 mM, KCl 4.9 mM, $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ 1.46 mM, MgSO_4 1.2 mM, KH_2PO_4 1.2 mM, NaHCO_3 25 mM, d-glucose 11.1 mM; pH 7.4)¹¹ and maintained at 37°C with continuously bubbled with a gas mixture of 95% O_2 /5% CO_2 . All chemicals used for Krebs solution were obtained from Sigma-Aldrich (St. Louis, MO). Reactivity of the isolated vascular rings was measured and the tension was recorded using isometric force transducers (MLT0420, ADInstruments) connected to an acquisition system (PowerLab 16/35, ADInstruments) coupled to a recording software (LabChart Data Analysis Software, ADInstruments).

The aortic rings were stabilized under a resting tension of 1.0 g for 60 minutes before measurement. Then the aortic rings were exposed to one of the vasoconstrictors, phenylephrine (1 nM to 100 µM; Sigma-Aldrich) or angiotensin II (1 nM to 100 µM; Sigma-Aldrich), to induce contraction and the contraction obtained was recorded. Then the vasodilator acetylcholine (1 mM; Sigma-Aldrich) was added to induce relaxation. To ensure the integrity of endothelium, only those aortic ring preparations that acetylcholine could induce at least 80% relaxation in vasoconstrictor-induced contraction were used in this study. Each preparation was exposed to only one vasoconstrictor (either phenylephrine or angiotensin II). With the completion of each cycle of contraction and relaxation, the preparations were washed with Krebs solution and were stabilized for 30 min before initiation of the next cycle of contraction and relaxation. Cumulative concentration response curves to vasoconstrictors were then constructed for vascular reactivity analysis.

Inflammatory mediator prostaglandin E_2 measurement

Snap frozen thoracic aorta tissues were processed as previously reported.¹² Concentration of prostaglandin E_2 (PGE_2) of the supernatant of thoracic aorta was assayed using

enzyme-linked immunosorbent assay. A commercial available PGE₂ ELISA Kit (Pierce Biotechnology Inc, Rockford, IL) was employed and the assay was performed according to the manufacturer's guidelines.

Cyclooxygenase-2 translational expression measurement

Immunoblotting assay was conducted to evaluate translational expression of cyclooxygenase-2 (COX-2), the enzyme regulating PGE₂ production,¹³ according to previously published protocols.^{12,14} In brief, proteins from snap frozen thoracic aorta tissues were extracted and the concentrations were measured. Equal amounts of protein (65 µg) were loaded to each well of 7.5% tris-glycine polyacrylamide gels and then were separated by electrophoresis followed by transferal to nitrocellulose membranes (Bio-Rad Laboratories, Hercules, CA). Membranes were incubated in primary antibody solution of COX-2 (1:1000 dilution, polyclonal COX-2 antibody, Cell Signaling Technology, Inc, Danvers, MA) or actin (internal standard; 1:1500 dilution, monoclonal β-actin antibody, Cell Signaling Technology, Inc, Danvers, MA) overnight at 4°C. After washing, membranes were then incubated in secondary antibody solution (horseradish peroxidase conjugated antimouse antibody, 1:1000 dilution, Amersham Pharmacia Biotech Inc, Piscataway, NJ) for 1 hour at room temperature. Bound antibody was detected, and the band densities were measured (Scion Corporation, Frederic, MD).

Assays of nitric oxide and malondialdehyde for oxidative status evaluation

Snap frozen thoracic aorta tissues were processed as above-mentioned. Then, the concentrations of nitric oxide (NO) (reactive nitrogen species)¹⁵ and malondialdehyde (MDA) (an indicator of lipid peroxidation)¹⁶ of the supernatants of thoracic aorta were measured. Commercially available colorimetric assay kits (Nitrate/Nitrite Colorimetric Assay Kit, Cayman Chemical, Ann Arbor, MI; Lipid Peroxidation Assay Kit, Oxford Biomedical Research, Oxford, MI) were employed and assays were performed according to the manufacturers' guidelines.

Statistical analysis

Two-way analysis of variance followed by Dunnett's multiple comparison test was performed to compare the between-group difference. All data were presented as means ± standard deviations. The significance level was set as 0.05. The SPSS 16.0 (IBM, Armonk, NY) was used for statistical analyses.

Results

The number of surviving rats before euthanization of the Sham, the P, the CLP, the CLP + P(10), the CLP + P(50), and the CLP + P(100) groups was 6, 6, 5, 5, 6, and 6, respectively.

CLP decreased vascular reactivity induced by vasoconstrictors

Figure 1 illustrated the representative contraction trace of the aortic rings after exposure to vasoconstrictors. Our data revealed that both vasoconstrictors phenylephrine and angiotensin II induced significant contractions in aortic rings of the Sham and the P groups (Fig. 2A and B). The maximal contractile effects (E_{max} , in grams) and the vascular reactivity induced by phenylephrine and angiotensin II of the Sham and the P groups were comparable (all $P > 0.05$; Table, Fig. 2A and B). Of note, phenylephrine and angiotensin II also induced significant contractions in aortic rings of the CLP group (Fig. 2A and B). However, the E_{max} induced by phenylephrine and angiotensin II of the CLP group were significantly lower than those of the Sham group (both $P < 0.001$, Table). In addition, the vascular reactivity induced by phenylephrine and angiotensin II of the CLP group were also significantly lower than those of the Sham group (both $P < 0.001$; Fig. 2A and B).

Platonin mitigated the effects of CLP on inhibiting vascular reactivity.

The E_{max} and the vascular reactivity induced by phenylephrine and angiotensin II of the CLP and the CLP + P(10) groups were not significantly different (all $P > 0.05$; Table, Fig. 3A and B). In contrast, the E_{max} induced by phenylephrine and angiotensin II of the CLP + P(50) and the CLP + P(100) groups were significantly higher than those of the CLP group (phenylephrine: $P = 0.003$ and < 0.001 ; angiotensin II: $P = 0.004$ and < 0.001 ; Table). The vascular reactivity induced by phenylephrine and angiotensin II of the CLP + P(50) and the CLP + P(100) groups were also significantly higher than those of the CLP group (phenylephrine: $P = 0.024$ and 0.017 , Figure 3A; angiotensin II: $P = 0.031$ and 0.036 , Fig. 3B).

Platonin mitigated the effects of CLP on upregulating PGE₂/COX-2 expression.

PGE₂ and COX-2 concentrations of thoracic aorta of the Sham and the P groups were both low (Fig. 4A and B). PGE₂ and COX-2 concentrations of thoracic aorta of the CLP group were significantly higher than those of the Sham group (both $P < 0.001$). PGE₂ and COX-2 concentrations of thoracic aorta of the CLP and the CLP + P(10) groups were not significantly different (both $P > 0.05$; Fig. 4A and B). Of note, PGE₂ and COX-2 concentrations of the CLP group were significantly higher than those of the CLP + P(50) and the CLP + P(100) groups

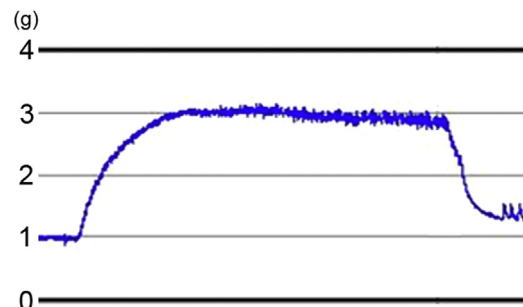


Fig. 1 – The representative contraction trace of the thoracic aortic rings after exposure to vasoconstrictors. (Color version of figure is available online.)

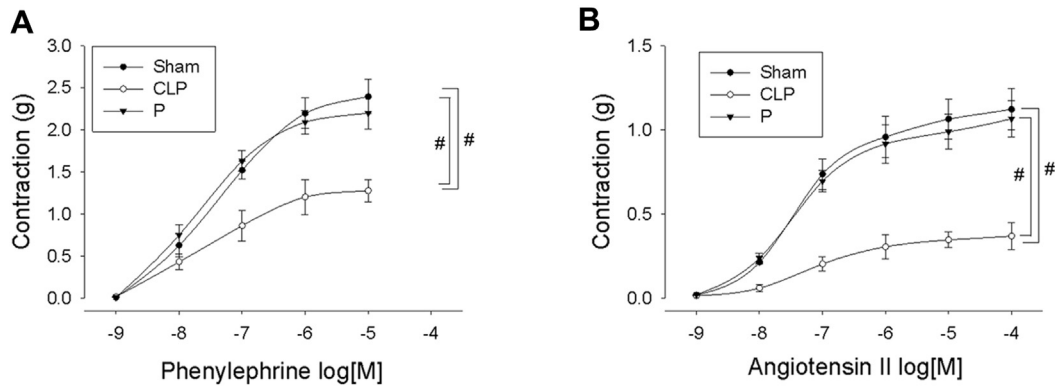


Fig. 2 – The contractile effects of isolated thoracic aortic rings induced by (A) phenylephrine and (B) angiotensin II in rats of the sham operation (Sham), the Sham plus 100 $\mu\text{g/kg}$ platonin (P) and the cecal ligation and puncture (CLP) groups. Data are mean $\pm$ standard deviations. $^{\#}P < 0.05$ versus the CLP group.

(PGE₂: $P = 0.034$ and 0.012 , Figure 4A; COX-2: $P = 0.021$ and 0.005 , Fig. 4B).

Platonin mitigated the effects of CLP on increasing oxidation

NO and MDA concentrations of thoracic aorta were both low in the Sham and the P groups (Fig. 5A and B). As expected, NO and MDA concentration of the CLP group were significantly higher than those of the Sham group (both $P < 0.001$). NO and MDA concentration of thoracic aorta of the CLP group were also significantly higher than those of the CLP + P(50) and the CLP + P(100) groups (NO: $P = 0.018$ and 0.003 , Figure 5A; MDA: $P = 0.037$ and 0.014 , Fig. 4B), but not the CLP + P(10) group (both $P > 0.05$; Fig. 5A and B).

Discussion

Data from the present study revealed that the maximal contractile effects E_{max} and vascular reactivity of thoracic aortic rings induced by vasoconstrictors in rats receiving CLP were significantly lower than those in rats receiving sham operations. These data confirmed that the CLP polymicrobial sepsis model employed in this study could readily induce significant vascular hyporeactivity in rats. Our data also revealed that the E_{max} and vascular reactivity of aortic rings induced by vasoconstrictors in rats receiving CLP plus platonin (50 and 100 $\mu\text{g/kg}$) were significantly higher than those in rats receiving CLP. These data confirmed our hypothesis and

demonstrated that platonin could attenuate the effects of polymicrobial sepsis and mitigate vascular hyporeactivity in rats receiving CLP. In a previous report, platonin was shown to mitigate circulatory failure in septic rats.³ Data from this study thus provide clear evidence to further depict the mechanism underlying the abovementioned beneficial effects of platonin against sepsis.³ As platonin is currently used in clinical conditions,² impact of our data should be profound and warrant further investigation.

As aforementioned, inflammation and oxidative stress are two crucial mechanisms underlying sepsis-induced vascular hyporeactivity.⁸ Previous data have confirmed the antiinflammation effects of platonin.¹⁷⁻¹⁹ For instance, we have previously confirmed the effects of platonin on decreasing inflammatory cytokines upregulation induced by endotoxin.¹⁷ Previous data further highlighted that the potent antiinflammation effects platonin may involve its effects on inhibiting the upstream regulatory pathways, including nuclear factor- κB and mitogen-activated protein kinases.^{18,19} In addition, platonin is a photosensitive dye that previous studies have confirmed its potent antioxidation effects.^{20,21} Among the identified mechanisms, direct free radical-scavenging effect is one the most significant mechanisms that account for the potent antioxidation effect of platonin.^{20,21} The antioxidation effect was further confirmed by our previous reports.^{5,6,22} Using rodent models of hemorrhage/resuscitation and lower limb ischemia-reperfusion, we have previously confirmed that platonin could exert significant antioxidation effect and attenuate organ injuries induced by

Table – The maximal contractile effects (E_{max} , in grams) in thoracic aortic rings at 12 hours after interventions.

	Sham (n = 6)	P (n = 6)	CLP (n = 5)	CLP + P(10) (n = 5)	CLP + P(50) (n = 6)	CLP + P(100) (n = 6)
Phenylephrine (0.1 μM)	2.54 $\pm$ 0.34	2.35 $\pm$ 0.21 [†]	1.33 $\pm$ 0.19 [*]	1.44 $\pm$ 0.22 [*]	2.02 $\pm$ 0.32 ^{*,†}	2.12 $\pm$ 0.30 ^{*,†}
Angiotensin II (0.1 μM)	1.12 $\pm$ 0.26	1.07 $\pm$ 0.21 [†]	0.37 $\pm$ 0.08 [*]	0.43 $\pm$ 0.07 [*]	0.76 $\pm$ 0.10 ^{*,†}	0.86 $\pm$ 0.11 ^{*,†}

^{*} $P < 0.05$ versus the Sham group.

[†] $P < 0.05$ versus the CLP group.

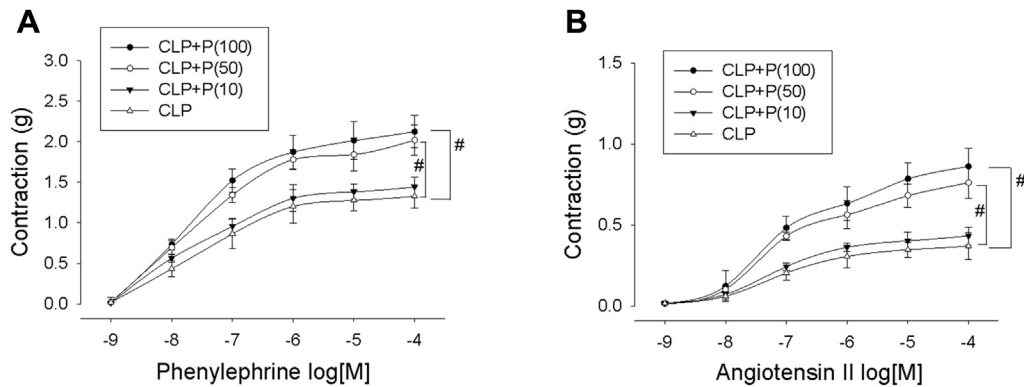


Fig. 3 – The contractile effects of isolated thoracic aortic rings induced by (A) phenylephrine and (B) angiotensin II in rats of the cecal ligation and puncture (CLP), the CLP plus 10 $\mu\text{g/kg}$ platonin (P) [CLP + P(10)], CLP plus 50 $\mu\text{g/kg}$ platonin [CLP + P(50)] and the CLP plus 100 $\mu\text{g/kg}$ platonin [CLP + P(100)] groups. Data are mean $\pm$ standard deviations. $^{\#}P < 0.05$ versus the CLP group.

hemorrhage/resuscitation⁵ and ischemia-reperfusion.^{6,22} In line with this notion, we thus speculate that the mechanisms underlying the effects of platonin on mitigating sepsis-induced vascular hyporeactivity may very likely involve its antiinflammation and antioxidation effects. This concept is supported by our data that platonin could inhibit upregulation of inflammatory mediator and oxidation of thoracic aorta induced by CLP polymicrobial sepsis.

Certain study limitations do exist and need to be acknowledged. First, the effects of platonin were examined at the early phase of sepsis (i.e. 12 hours after CLP) in this study. The question of whether platonin can exert similar effects in the delay phase of sepsis remains unstudied. Second, our data revealed that platonin at the dosages of 50 and 100 $\mu\text{g/kg}$ exerted beneficial effects on restoring vascular reactivity in septic rats. However, more studies are needed before we can conclude on the overall effects of platonin with different dosages.

In conclusions, data from this rodent study confirmed that platonin can mitigate vascular hyporeactivity induced by polymicrobial sepsis.

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Authors' contributions: T.C.P., C.Y.C., and C.J.H. contributed to the conception and design, data collection, analysis and interpretation, writing, and critical revision of the article. I.T.H., M.C.K., and Y.Y.C. contributed to data analysis and interpretation as well as writing and critical revision of the article.

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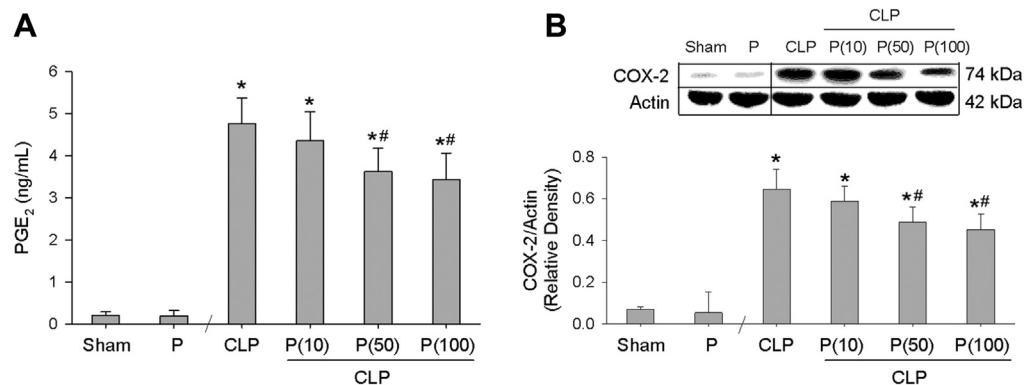


Fig. 4 – (A) The concentrations of prostaglandin E₂ (PGE₂) and (B) representative gel photography and densitometric analysis data illustrate the translational expression of cyclooxygenase-2 (COX-2) of thoracic aorta in rats of the sham operated (Sham), the Sham plus 100 $\mu\text{g/kg}$ platonin (P), the cecal ligation and puncture (CLP), the CLP plus 10 $\mu\text{g/kg}$ platonin [CLP + P(10)], CLP plus 50 $\mu\text{g/kg}$ platonin [CLP + P(50)] and the CLP plus 100 $\mu\text{g/kg}$ platonin [CLP + P(100)] groups. Actin was used as the internal standard. Data are mean $\pm$ standard deviations. $^*P < 0.05$ versus the Sham group. $^{\#}P < 0.05$ versus the CLP group.

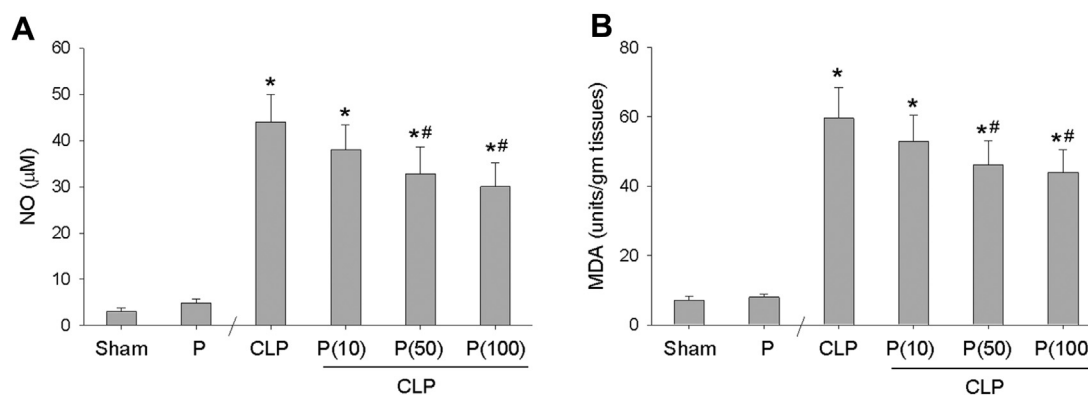


Fig. 5 – The concentrations of (A) nitric oxide (NO) and (B) malondialdehyde (MDA) of thoracic aorta in rats of the sham operated (Sham), the Sham plus 100 $\mu\text{g/kg}$ platonin (P), the cecal ligation and puncture (CLP), the CLP plus 10 $\mu\text{g/kg}$ platonin [CLP + P(10)], CLP plus 50 $\mu\text{g/kg}$ platonin [CLP + P(50)] and the CLP plus 100 $\mu\text{g/kg}$ platonin [CLP + P(100)] groups. Data are mean $\pm$ standard deviations. * $P < 0.05$ versus the Sham group. # $P < 0.05$ versus the CLP group.

Disclosure

The authors report no proprietary or commercial interest in any product mentioned or concept discussed in this article.

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